

Generation Interconnection Request System Impact Study Report

For Queue Project Q16-0634

Salt Fork Wind 138 kV
November 15, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0634. The Interconnection Customer proposes a 56.2 MW solar photovoltaic facility interconnecting at the Salt Fork Wind 138 kV substation in the Kentucky area. The request was received on June 20, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 268 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,710,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: WISSOTA-BERRY ST 138 kV
Contingency: DAVISVILLE-NORTH APPLETON 138 kV
Loading: 117.3% (Rating: 421 MVA)
DFAX: 11.2%
Pre-project loading: 115.8%
2. Facility: GUADALUPE-MISTERSKY 138 kV L
Contingency: WISSOTA-BERRY STREET 138 kV
Loading: 114.6% (Rating: 174 MVA)
DFAX: 6.4%
Pre-project loading: 112.6%
3. Facility: SAND SPRINGS-OJO 138 kV
Contingency: NORTHFIELD-BERRY STREET 345 kV
Loading: 112.8% (Rating: 322 MVA)
DFAX: 5.8%
Pre-project loading: 111.8%
4. Facility: WELLINGTON-DAVISVILLE 138 kV
Contingency: KILLEN-TOLEDO BEND 500 kV L
Loading: 109.4% (Rating: 315 MVA)
DFAX: 5.1%
Pre-project loading: 108.5%
5. Facility: CLAUS-SAN ANDRES 138 kV
Contingency: KILLEN-TOLEDO BEND SUBSTATION 500 kV
Loading: 109.2% (Rating: 249 MVA)
DFAX: 6.5%
Pre-project loading: 107.7%

6. Facility: SAN ANDREAS-WISSOTA 138 kV Line
Contingency: TOLEDO BEND-KILLEN 500 kV
Loading: 109.0% (Rating: 253 MVA)
DFAX: 6.5%
Pre-project loading: 107.5%
7. Facility: SAND SPRINGS-SNOHOMISH 138 kV line
Contingency: WISSOTA-BERRY ST 138 kV line
Loading: 108.7% (Rating: 314 MVA)
DFAX: 5.0%
Pre-project loading: 107.8%
8. Facility: MORTAN-GUADALUPE 138 kV line
Contingency: WISSOTA-BERRY STREET 138 kV line
Loading: 108.6% (Rating: 201 MVA)
DFAX: 6.4%
Pre-project loading: 106.8%
9. Facility: MAINWARING-MORTAN 138 kV
Contingency: BERRY STREET SUBSTATION-WISSOTA 138 kV
Loading: 108.2% (Rating: 204 MVA)
DFAX: 6.4%
Pre-project loading: 106.5%
10. Facility: UNION-CLAUS 138 kV
Contingency: TOLEDO BEND-KILLEN 500 kV
Loading: 106.6% (Rating: 236 MVA)
DFAX: 6.5%
Pre-project loading: 105.1%
11. Facility: MOFFAT 345 kV/230 kV Autotransformer
Contingency: MOFFAT-KILLEN 500 kV
Loading: 105.6% (Rating: 456 MVA)
DFAX: 7.9%
Pre-project loading: 104.7%
12. Facility: LOVELAND-OJO 138 kV
Contingency: NORTHFIELD-BERRY ST 345 kV
Loading: 104.5% (Rating: 223 MVA)
DFAX: 5.8%
Pre-project loading: 103.1%
13. Facility: WASHINGTON COUNTY COGENERATION FACILITY-MOSSYROCK 230 kV
Contingency: WOLFE-BARRETT 500 kV Line
Loading: 103.5% (Rating: 739 MVA)
DFAX: 9.8%
Pre-project loading: 102.8%
14. Facility: MOFFAT 345/230kV Auto Bank 1
Contingency: MOFFAT-KILLEN 500 kV
Loading: 103.2% (Rating: 467 MVA)
DFAX: 7.9%
Pre-project loading: 102.2%
15. Facility: LOVELAND-BERRY ST 138 kV
Contingency: BERRY ST-NORTHFIELD 345 kV
Loading: 102.2% (Rating: 224 MVA)
DFAX: 5.8%
Pre-project loading: 100.8%
16. Facility: NARCOOSSEE-PICNIC POINT 100 kV Circuit 1
Contingency: NARCOOSSEE-PICNIC PT 100 kV Line 2

Loading: 101.6% (Rating: 153 MVA)

DFAX: 8.6%

Pre-project loading: 98.5%

17. Facility: NARCOOSSEE-PICNIC PT 100 kV

Contingency: NARCOOSSEE-PICNIC POINT 100 kV #1

Loading: 101.6% (Rating: 153 MVA)

DFAX: 8.6%

Pre-project loading: 98.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WISSOTA-BERRY ST 138 kV	Rebuild line with higher-capacity conductor	\$970,000	2% (\$24,000)
MISTERSKY-GUADAUPE 138 kV	Rebuild line with higher-capacity conductor	\$14,081,000	3% (\$367,000)
OJO-SAND SPRINGS 138 kV	Rebuild line with higher-capacity conductor	\$6,925,000	2% (\$140,000)
WELLINGTON-DAVISVILLE 138 kV	Reconductor line	\$1,321,000	2% (\$29,000)
SAN ANDREAS-CLAUS 138 kV	Reconductor line	\$3,950,000	3% (\$109,000)
WISSOTA-SAN ANDREAS 138 kV L	Reconductor line	\$2,315,000	3% (\$64,000)
SAND SPRINGS-SNOHOMISH 138 kV line	Reconductor line	\$5,414,000	2% (\$115,000)
GUADALUPE-MORTAN 138 kV	Reconductor line	\$4,746,000	3% (\$124,000)
MAINWARING-MORTAN 138 kV	Reconductor line	\$3,373,000	3% (\$88,000)
CLAUS SUBSTATION-UNION 138 kV	Reconductor line	\$654,000	3% (\$18,000)
MOFFAT 345 kV/230 kV Autotransformer	Replace transformer with higher-rated unit	\$15,864,000	3% (\$506,000)
LOVELAND-OJO 138 kV	Reconductor line	\$1,691,000	2% (\$41,000)
WASHINGTON COUNTY COGENERATION FACILITY-MOSSYROCK 230 kV	Reconductor line	\$17,906,000	1% (\$253,000)
MOFFT 345-230 kV XFMR Unit 1	Replace transformer with higher-rated unit	\$21,694,000	3% (\$692,000)
BERRY STREET-LOVELAND 138 kV	Reconductor line	\$2,763,000	2% (\$67,000)
NARCOOSSEE-PICNIC POINT 100 kV Circuit 1	Reconductor line	\$390,000	7% (\$27,000)
NARCOOSSEE-PICNIC PT 100 kV	Reconductor line	\$668,000	7% (\$46,000)
	Total allocated to project		\$2,710,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.